

Topology Munkres - Solutions Manual

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Chapter 1

Set Theory and Logic

1.1 Fundamental Concepts

Exercise 1

Okay.

Exercise 2

- (a) Only $\Rightarrow$ holds.
- (b) Only $\Rightarrow$ holds.
- (c) True.
- (d) Only $\Leftarrow$ holds.
- (e) Only $\subseteq$ holds.
- (f) Only $\supseteq$ holds.
- (g) True:

$$\begin{aligned} & A \cap (B - C) \\ &= A \cap (B \cap C^c) \\ &= (A \cap B) \cap C^c \\ &= \emptyset \cup ((A \cap B) \cap C^c) \\ &= ((A \cap B) \cap A^c) \cup ((A \cap B) \cap C^c) \\ &= (A \cap B) \cap (A^c \cup C^c) \\ &= (A \cap B) \cap (A \cap C)^c \\ &= (A \cap B) - (A \cap C) \end{aligned}$$

(h+) Too lazy.

Exercise 3

(a) Contrapositive:
If $x^2 - x \leq 0$, then $x \geq 0$.

Converse:

If $x^2 - x > 0$, then $x < 0$.

Only the contrapositive is true.

(b) Contrapositive:
If $x^2 - x \leq 0$, then $x \leq 0$.

Converse:

If $x^2 - x > 0$, then $x > 0$.

None of them are true.

Exercise 4

- (a) There exists $a \in A$ such that $a^2 \notin B$.
- (b) For all $a \in A$, we have $a^2 \notin B$.
- (c) There exists $a \in A$ such that $a^2 \in B$.
- (d) For all $a \notin A$, we have $a^2 \notin B$.

Exercise 5

- (a) Both statement and converse are true by definition.
- (b) Statement is false. Converse is true.
- (c) Statement is true. Converse is false.
- (d) Both statement and converse are true by definition.

Exercise 6

Too lazy.

Exercise 7

$$D = A \cap (B \cup C)$$

$$E = (A \cap B) \cup C$$

$$F = A - (B - C)$$

Exercise 8

If A has n elements, then $\mathcal{P}(A)$ has 2^n elements.

Exercise 9

Done this so many times. In other textbooks.

Exercise 10

- (a) Yes. $\mathbb{Z} \times \mathbb{R}$
- (b) Yes. $\mathbb{R} \times (0, 1]$
- (c) No.
- (d) Yes. $(\mathbb{R} - \mathbb{Z}) \times \mathbb{Z}$
- (e) No.

1.2 Functions**Exercise 1**

- (a) For any function, we have

$$x \in A_0 \Rightarrow f(x) \in f(A_0) \Rightarrow x \in f^{-1}(f(A_0))$$

Now, let f injective. By definition,

$$x \in f^{-1}(f(A_0)) \Rightarrow f(x) \in f(A_0)$$

Suppose for contradiction that $x \notin A_0$. Denote $y = f(x)$. We know y is an element of $f(A_0)$. In other words, some element of A_0 must be able to create it, i.e. $y = f(z)$ for some $z \in A_0$. As we said, since it's not x , then there must exist some other $z \in A_0$ where $z \neq x$, such that $f(z) = y$. However, this contradicts the fact that f is injective, i.e. we have $z \neq x \Rightarrow f(z) \neq f(x)$. Hence, $x \in A_0$ is true.

(b) If $y \in f(f^{-1}(B_0))$, then $y = f(x)$ for some $x \in f^{-1}(B_0)$. By definition, the latter means that $f(x) \in B_0$. We know $y = f(x)$, hence $y \in B_0$.

Now, let f be surjective. If $y \in B_0$, then by surjectivity, we have that $f^{-1}(\{y\})$ is non-empty. And by definition, for every element $x \in f^{-1}(\{y\})$, we have $f(x) = y \in B_0$. Hence, $\{y\} \subseteq f(f^{-1}(\{y\}))$, thus $B_0 \subseteq f(f^{-1}(B_0))$.

Exercise 2

- (a)

$$x \in f^{-1}(B_0) \Rightarrow f(x) \in B_0 \Rightarrow f(x) \in B_1 \Rightarrow x \in f^{-1}(B_1)$$

(The second " $\Rightarrow$ " above uses the assumption given.)

- (b)

$$x \in f^{-1}(B_0 \cup B_1) \Rightarrow f(x) \in B_0 \cup B_1$$

If $f(x) \in B_0$, then $x \in f^{-1}(B_0)$. Likewise, if $f(x) \in B_1$, then $x \in f^{-1}(B_1)$. Hence, $x \in f^{-1}(B_0) \cup f^{-1}(B_1)$.

For the reverse direction, suppose that

$$x \in f^{-1}(B_0) \cup f^{-1}(B_1)$$

If $x \in f^{-1}(B_0)$, then $x \in f^{-1}(B_0 \cup B_1)$. Likewise, if $x \in f^{-1}(B_1)$, then $x \in f^{-1}(B_0 \cup B_1)$. Hence, $x \in f^{-1}(B_0 \cup B_1)$.

(c)

$$\begin{aligned}
& x \in f^{-1}(B_0 \cap B_1) \\
& \Leftrightarrow f(x) \in B_0 \cap B_1 \\
& \Leftrightarrow f(x) \in B_0 \text{ and } f(x) \in B_1 \\
& \Leftrightarrow x \in f^{-1}(B_0) \text{ and } x \in f^{-1}(B_1) \\
& \Leftrightarrow x \in f^{-1}(B_0) \cap f^{-1}(B_1)
\end{aligned}$$

(d)

$$\begin{aligned}
& x \in f^{-1}(B_0 - B_1) \\
& \Leftrightarrow x \in f^{-1}(B_0 \cap B_1^c) \\
& \Leftrightarrow f(x) \in B_0 \cap B_1^c \\
& \Leftrightarrow f(x) \in B_0 \text{ and } f(x) \notin B_1 \\
& \Leftrightarrow x \in f^{-1}(B_0) \text{ and } x \notin f^{-1}(B_1) \\
& \Leftrightarrow x \in f^{-1}(B_0) - f^{-1}(B_1)
\end{aligned}$$

(e)

$$y \in f(A_0) \Rightarrow y = f(x) \mid x \in A_0 \Rightarrow y = f(x) \mid x \in A_1 \Rightarrow y \in f(A_1)$$

(The second " $\Rightarrow$ " above uses the assumption given.)

(f)

$$y \in f(A_0 \cup A_1) \Rightarrow y = f(x) \mid x \in A_0 \cup A_1$$

If $x \in A_0$, then $y \in f(A_0)$. Otherwise, $y \in f(A_1)$. Hence, $y \in f(A_0) \cup f(A_1)$.

(g)

$$\begin{aligned}
& y \in f(A_0 \cap A_1) \\
& \Rightarrow y = f(x) \mid x \in A_0 \cap A_1 \\
& \Rightarrow y = f(x) \mid x \in A_0 \text{ and } y = f(x) \mid x \in A_1 \\
& \Rightarrow y \in f(A_0) \text{ and } y \in f(A_1) \\
& \Rightarrow y \in f(A_0) \cap f(A_1)
\end{aligned}$$

Now, suppose that f is injective.

$$\begin{aligned}
& y \in f(A_0) \cap f(A_1) \\
& \Rightarrow y \in f(A_0) \text{ and } y \in f(A_1) \\
& \Rightarrow y = f(x_0) \mid x_0 \in A_0 \text{ and } y = f(x_1) \mid x_1 \in A_1
\end{aligned}$$

By injectivity, $x_0 = x_1$, thus

$$\begin{aligned}
& \Rightarrow y = f(x) \mid x \in A_0 \text{ and } x \in A_1 \\
& \Rightarrow y = f(x) \mid x \in A_0 \cap A_1
\end{aligned}$$

$$\Rightarrow y \in f(A_0 \cap A_1)$$

(h)

$$y \in f(A_0) - f(A_1)$$

$$\Rightarrow y \in f(A_0) \cap f^c(A_1)$$

$$\Rightarrow y \in f(A_0) \text{ and } y \notin f(A_1)$$

$$\Rightarrow y = f(x_0) \mid x_0 \in A_0 \text{ and } y \neq f(x_1) \mid \forall x_1 \in A_1$$

Since $y \in \text{range } f|_{A_0}$, there must exist $x \in A_0$ but $x \notin A_1$ such that $f(x) = y$.

$$\Rightarrow y = f(x) \mid x \in A_0 - A_1$$

$$\Rightarrow y \in f(A_0 - A_1)$$

Now, suppose that f is injective.

$$y \in f(A_0 - A_1)$$

$$\Rightarrow y = f(x) \mid x \in A_0 \cap A_1^c$$

We already know, by conjunction elimination, that $y = f(x) \mid x \in A_0$, hence $y \in f(A_0)$.

Now, for the second part, since $x \in A_1^c$, then $x \notin A_1$. Suppose for contradiction that $y \in f(A_1)$. Since $y \in f(A_1)$, there must exist $x' \in A_1$ such that $f(x') = y$. We know $x' \in A_1$ and $x \notin A_1$, hence $x' \neq x$. However, $f(x') = y = f(x)$. This contradicts the injectivity of f . Hence, it must be that $y \notin f(A_1)$.

Thus, $y \in f(A_0)$ and $y \notin f(A_1)$. Hence,

$$\Rightarrow y \in f(A_0) - f(A_1)$$

If you didn't understand, read the following. The crux to understanding the second part is to realize that y is a value in the codomain. Many x can produce this single y . We need to verify that every single $x \in f^{-1}(\{y\})$ is not contained in A_1 . If even a single one is found to be in A_1 , then the entire y is contained in $f(A_1)$. That's where injectivity comes in, guaranteeing that the preimage is a singleton, so that, as soon as we find a particular $x \notin A_1$ such that $f(x) = y$, then there is no more x to verify.

Exercise 3

Hell nah.

Exercise 4

(a)

$$\begin{aligned}
(g \circ f)^{-1}(C_0) &= \{a \in A \mid (g \circ f)(a) \in C_0\} \\
&= \{a \in A \mid \exists b \in B : f(a) = b \text{ and } g(b) \in C_0\} \\
&= \{a \in A \mid \exists b \in B : f(a) = b \text{ and } b \in g^{-1}(C_0)\} \\
&= \{a \in A \mid f(a) \in g^{-1}(C_0)\} \\
&= f^{-1}(g^{-1}(C_0))
\end{aligned}$$

(b)

$$x \neq x' \Rightarrow f(x) \neq f(x') \Rightarrow g(f(x)) \neq g(f(x'))$$

(c) Recall that Munkres uses the word "range" to mean "codomain". Hence, we only require f to be injective. g must be injective on $f(A) \subseteq B$. But it can behave freely elsewhere, i.e. on $B - f(A)$, because these values are never reached by f .

(d) For every $c \in C$, we can find $b \in B$ such that $g(b) = c$. And for every $b \in B$, we can find $a \in A$ such that $f(a) = b$. Thus, for every $c \in C$, we can find $a \in A$ such that $c = g(f(a))$. Hence, $g \circ f$ is surjective.

(e) g must be surjective. However, f need not be surjective, because it's possible that $C = g(B_0)$ for some subset $B_0 \subset B$. In other words, f only needs to reach B_0 for the entire composite to be surjective.

(f) Too lazy.

Exercise 5

(a) Let $f : A \rightarrow B$ have a left inverse $g : B \rightarrow A$ where $g \circ f = i_A$. Suppose for contradiction that f is not injective, i.e. there exists $x \neq x'$ such that $f(x) = f(x')$. Then,

$$g(f(x)) = g(f(x'))$$

But $g \circ f$ is the identity function, hence

$$\Rightarrow x = x'$$

contradicting our assumption. Hence, f is injective.

Now, let $f : A \rightarrow B$ have a right inverse $h : B \rightarrow A$ where $f \circ h = i_B$. For every $b \in B$, we can compute $h(b) \in A$. The identity function tells us that $f(h(b)) = b$. In other words, for every $b \in B$, we found $a \in A$ such that $f(a) = b$, where $a = h(b)$. This is precisely the requirement of surjectivity of f .

(b, c) It's really just playing with the codomain.

(d) Yes. Again, when f is not surjective, f need not map to the entirety of B , so g is free to do whatever it likes on the remaining portions. And when f is not injective, it can have multiple right inverses. For example, if $x, x' \in A$ both map to $y \in B$, then $h(b)$ can choose between them.

(e) Well, since f is both injective and surjective, then it's bijective.

First, h is surjective. Suppose for contradiction that it's not surjective. Then take any $x \in A - h(B)$, which is non-empty by non-surjectivity of h . Every $y \in B$ has already been taken by some $x' \in h(B)$, because $f(h(B)) = B$ by property of identity. Since x is in the domain of f , then it must produce something under f , say $y \in B$. But then $f(x) = y = f(x')$. We know $x' \neq x$, because they belong in complementary subsets of A . This would contradict the injectivity of f , hence h is surjective.

Now we can complete the proof. Given $y \in B$, using the surjectivity of f , there exists $x \in A$ such that

$$y = f(x)$$

Hence,

$$g(y) = g(f(x)) = x$$

by property of identity. Now, since h is surjective, then there must exist some $y' \in B$ such that

$$h(y') = x$$

It follows that

$$\begin{aligned} f(h(y')) &= f(x) \\ \Rightarrow y' &= y \end{aligned}$$

Hence, $h(y) = h(y') = x = g(y)$ for all $y \in B$, and since they share this same domain, then they are identical functions.

By injectivity of f , the x that gives $y = f(x)$ is unique. Hence, $g = h = f^{-1}$ is the inverse function, since it evaluates to this unique x .

Exercise 6

Too lazy.

1.3 Relations

Exercise 1

Equality is reflexive, symmetric, and transitive. The equivalence classes can be seen if we fix $y - x^2 = c$. Which becomes $y = x^2 + c$. These are parabolas. The equivalence classes differ by the amount shifted vertically.

Exercise 2

We are basically removing elements from an already related set, and we have to prove that the remaining connections still form an equivalence relation.

For every element $a \in A$, we have $(a, a) \in C$. If $a \in A_0$, then $(a, a) \in A_0 \times A_0$, hence $(a, a) \in C \cap (A_0 \times A_0)$.

Suppose $(a, b) \in C \cap (A_0 \times A_0)$. Clearly, $(b, a) \in C$. And since $a \in A_0$ and $b \in A_0$, then $(b, a) \in A_0 \times A_0$. Thus, $(b, a) \in C \cap (A_0 \times A_0)$.

Suppose $(a, b), (b, c) \in C \cap (A_0 \times A_0)$. Clearly, $(a, c) \in C$. And since $a \in A_0$ and $c \in A_0$, then $(a, c) \in A_0$. Thus, $(a, c) \in C \cap (A_0 \times A_0)$.

Exercise 3

These are conditional statements. If no pair (a, b) exist, then we can't proceed. In other words, none of these statements enforce existence.

Exercise 4

- (a) Apply the properties of equality.
- (b) Clearly intuitive, but I'm confusing myself with proofs.

Exercise 5

- (a) $x - x = 0 \in \mathbb{Z}$, so it's reflexive. If $x - y = n \in \mathbb{Z}$, then $y - x = -n \in \mathbb{Z}$, so it's symmetric. If $x - y = n \in \mathbb{Z}$ and $y - z = m \in \mathbb{Z}$, then $x - z = x + (y - y) - z = (x - y) + (y - z) = n + m \in \mathbb{Z}$, so it's transitive.

Obviously, it's a superset of S , because

$$(x, y) \in S \Rightarrow y = x + 1 \Rightarrow y - x = 1 \in \mathbb{Z} \Rightarrow (x, y) \in S'$$

Numbers belonging to a same equivalence classes have the same mod 1.

- (b) If $a \in A$, then (a, a) is contained in every equivalence relation (because that's a property of equivalence relations). Hence, it's in their intersection.

If (a, b) is contained in the intersection, then it must be contained in each equivalence relation. And each equivalence relation must contain (b, a) , hence their intersection contains (b, a) as well.

If (a, b) and (b, c) is contained in the intersection, then they must be contained in each equivalence relation. And each equivalence relation must contain (a, c) , hence their intersection contains (a, c) as well.

- (c) $(x, x) \in T$ for every $x \in \mathbb{R}$. By the way, S is not an equivalence relation, because it's not symmetric, so since $(x, y) \in S$, then $(x, y) \in T$, and since T is an equivalence relation, then $(y, x) \in T$. In other words, we add the missing symmetry. Transitivity doesn't need to be dealt, because the range $0 < x < 2$ is too small for transitivity.

T basically looks like the real number line, but we glue the interval $(0, 1)$ with $(1, 2)$ sort of in a superposition fashion. There's a hole in the middle, with only $(1, 1) \in T$ (not interval, here I'm expressing a pair).

Exercise 6